

Review retrospective:
*the J -function quartic asymptotic, and a load-bearing open
mis-flagged again*

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2026-06-05

Abstract

A soundness review of `Laplace/OneD/JFunctionQuartic.lean`: the leading-order asymptotic of the J -function for the quartic potential, $t^{1/4} \int_{\mathbb{R}} e^{-t w^4/24} \pi(w) dw \rightarrow \pi(0) \cdot \frac{1}{2} \cdot 24^{1/4} \Gamma(1/4)$, via a substitute-then-dominated-convergence argument. Result: **a clean fidelity pass, no edits** — a genuinely substantive result (unlike the copy-paste quartic / sextic specialisations), with the one “dead open” flag rejected as a repeat of an earlier over-flag.

1 Setting

This is Step 2 of the grammar-paper precursor sequence: a single-component blow-up at the degenerate quartic. It consumes the pure-quartic moments (the partition closed form and the integrability) but the argument itself — a change of variables to a t -independent integrand followed by the dominated-convergence theorem — is new content, not a specialisation.

2 What was reviewed

Four declarations: a private substitution lemma ($u = t^{1/4}w$ reshapes the centred integral), a private dominator-integrability lemma, the headline centred DCT theorem (`quartic_jfunction_centered_tendsto_zero`: the prior-perturbation $\pi(w) - \pi(0)$ is negligible against the concentrating Gibbs weight), and the user-facing asymptotic corollary.

3 Fidelity / soundness findings

Sound; no mismatch. The substitution is correct ($a := t^{1/4}$, $a^4 = t$, $(wa)^4 = tw^4$, $(wa)/a = w$, $\int g(wa) dw = |a^{-1}| \int g$ with $|a^{-1}| = 1/a$ cancelling the outer a). The DCT step uses the right envelope $2M e^{-u^4/24}$ (from $|\pi(u/a) - \pi(0)| \leq 2M$) and the right pointwise limit ($u/t^{1/4} \rightarrow 0$, continuity, so $\pi(u/t^{1/4}) \rightarrow \pi(0)$). The corollary splits $\pi = (\pi - \pi(0)) + \pi(0)$, applies the partition closed form to the constant piece, and uses $t^{1/4}(24/t)^{1/4} = 24^{1/4}$ to land the constant $\pi(0) \cdot \frac{1}{2} \cdot 24^{1/4} \Gamma(1/4)$.

One report-only note: the theorems assume `Continuous prior` (global), while the docstring speaks of a prior “continuous (at 0)”. This is not a defect — global continuity is genuinely used to get `AEStronglyMeasurable` of the integrand (the pointwise limit alone needs only continuity at 0, but measurability of a merely-continuous-at-0 prior is not free). The docstring captures the

mathematical essence (continuity at 0 is what drives the limit); the global hypothesis is a sound formalisation choice. A forward tide could weaken it to `ContinuousAt + AEMeasurable`; it is not a review-tide edit.

4 Simplifications applied

None. The reviewer’s only structural suggestion — that `open Topology` “looks unused” — was **rejected**: the neighbourhood-filter notation $\mathcal{N}$ is the `Topology` scope’s, and it appears eight times in the file, *including the two public theorem statements* (`... atTop ( $\mathcal{N}$  0)`). Removing the open would break the build. This is the same over-flag the third-cumulant review caught earlier in the sweep — GPT mislabels a load-bearing `open Topology` because every $\mathcal{N}$ is written with the notation rather than a long-form name. The remaining suggestions (a duplicated $\leq 2M$ estimate, two helper `haves`) are tactic-golf and were declined. All three imports are genuinely used (including `Haar.NormedSpace`, for the change-of-variables `integral_comp_mul_right`); the file carries no cruft.

5 Disagreements and open questions

None on the mathematics. The hypothesis note is surfaced for the user, not escalated.

6 Lessons

- **The `open Topology/ $\mathcal{N}$` over-flag is now a recognised recurring false positive.** Twice in this sweep the reviewer has called `open Topology` dead while $\mathcal{N}$ — its scoped notation — was load-bearing, here even in the public signatures. The standing rule holds: a dead-open claim is checked by grepping for the *notation* the open provides ($\mathcal{N}$, $\int$, ...), not only for long-form names, and the build is the final arbiter. When $\mathcal{N}$ appears, `open Topology` stays.
- **Not every file in a churned area is cruft-laden.** After a run of specialisation files thick with copy-pasted dead imports and dead tactics, this substantive J-function result was clean — all imports used, all opens load-bearing, no warning. The audit value here was the fidelity confirmation of a non-trivial DCT argument, not a cleanup.

7 Follow-ups

- The sibling `JFunctionSextic.lean` and the more general `JFunctionMonomial.lean` / `JFunctionAsymptoticEquivalence.lean` are the natural continuations of the J-function arc; the `Multi/Covariance*` trio remains the large sectioned target.